# Personalized 'You may also like' on PDP

*Replacing the static 'frequently bought together' module with personalized recommendations on the product detail page increases add-to-cart rate and downstream conversion.*

AT A GLANCE

**Primary metric is *conversion_rate*; decision is binary against Ship if add_to_cart_rate lift exceeds MDE/2 (≥1% relative) with CI clearing zero.**

*Pre-registered before any analysis. Decision is binary against conversion_rate: a 20.0% relative relative lift over a 10.0% baseline requires 12,000 per arm at the standard a / power settings. Guardrails sit on revenue_per_user.*

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SAMPLE REQUIRED (PER ARM)

**12,000**

*to detect a 20.0% relative relative effect at a = 0.05, power = 0.80*

BASELINE
10.0%

DURATION
14 days

TOTAL SAMPLE
24,000

FEASIBILITY
VIABLE

MECHANISM

## Why we think this works

*Personalization based on past purchase history should surface more relevant complementary products than the static co-purchase baseline, especially for repeat customers.*

*The static module ships for everyone today; click-through on the module is ~6%. Personalization in the cart page (a separate test in 2025Q3) showed a 1.5pp lift in CTR.*

METRICS

## conversion_rate is the verdict-driving metric.

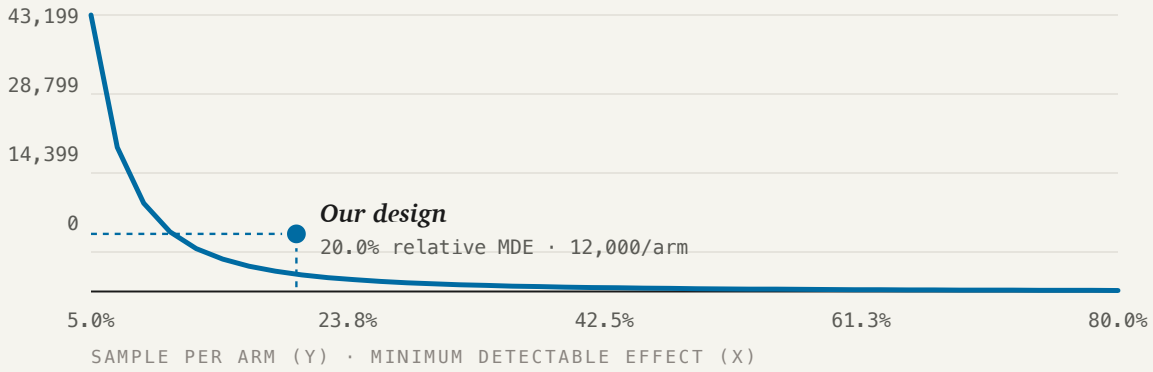
*Primary is conversion_rate; secondaries inform interpretation but do not influence the verdict. Guardrails block ship if their 90% CI lies past the pre-registered harm boundary.*

| | | |
|---|---|---|
| PRIMARY | **conversion_rate** | Proportion · baseline 10.0% · MDE 20.0% relative · direction higher is better |
| SECONDARY | **conversion_rate** | Leading indicator. Not used for the ship/no-ship decision but informs interpretation. |
| SECONDARY | **average_order_value** | Leading indicator. Not used for the ship/no-ship decision but informs interpretation. |
| GUARDRAIL | **revenue_per_user** | Direction higher is better. Decision blocks if 90% CI lies past -5.0% relative. |

POWER

## Detecting smaller effects requires much more sample.

*To resolve a 20.0% relative relative effect against a 10.0% baseline, we need 12,000 per arm. Tighter detection thresholds would climb the curve to the left — quickly.*




*Smaller effects require disproportionately more sample. Our design point is feasible inside the available user_id surface.*

ASSIGNMENT + COHORTS

## First-exposure attribution at the user_id level.

*Each user is randomly assigned at first exposure to control or treatment in a 50 / 50 split. Events before first_exposure_at are not attributed to the experiment.*

| | |
|---|---|
| UNIT | **user_id**<br>One row per (user_id, experiment_id) in the assignments table; first_exposure_at gates which events are attributed. |
| ARMS | **50 / 50**<br>control · treatment |
| COHORT | **all assigned users** |

DECISION RULES

## Pre-registered. The analyze verb walks the same tree against the same metrics.

*Ship if add_to_cart_rate lift exceeds MDE/2 (≥1% relative) with CI clearing zero. Smaller lifts surface a caveat.*

| | |
|---|---|
| SHIP | Primary 95% CI excludes zero on benefit side · no guardrail breach · novelty late-ratio ≥ 0.7 |
| LIFT WITH CAVEAT | Real lift but below MDE/2 — surface the caveat in the readout but still consider rolling out |
| NO-SHIP · GUARDRAIL | Any guardrail's 90% CI lies past its harm threshold |
| NO-LIFT | Well-powered, primary CI straddles zero with half-width > 2 × MDE |
| INCONCLUSIVE | Underpowered (n < required) AND primary CI straddles zero |
| INVALID | SRM $\chi^2$ $p < 0.0005$ and no override resolved — investigate assignment |